

SILVER BAY, MN, USA

WMO: 727556

Lat: 47.249N

Lon: 91.416W

Elev: 1089

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04905

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-14.7	-27.1	1.3	-17.6	-23.6	1.6	-11.8	18.0	21.0	16.5	19.9	1.9	290	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	84.1	68.5	81.6	66.1	79.1	64.4	71.3	80.9	68.9	77.8	66.9	75.2	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	107.2	77.2	65.7	98.7	74.9	63.6	91.6	71.1	35.7	80.6	33.7	78.4	32.1	75.3	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.6	12.8	DB	-28.4	90.4	6.8	3.3	-33.3	92.7	-37.2	94.6	-41.1	96.5	-46.0	98.8	(4)
(5)			WB	-28.3	75.0	6.6	2.2	-33.1	76.6	-37.0	77.8	-40.7	79.1	-45.5	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	10.9	13.9	24.8	36.6	48.3	58.1	64.4	62.4	54.6	42.4	28.9	16.5		
	DBStd	20.83	13.20	12.24	11.74	8.17	8.10	6.99	6.04	5.81	7.97	7.65	10.09	12.04		
	HDD50	5537	1213	1012	782	407	131	14	0	1	43	261	634	1039		
	HDD65	9772	1678	1432	1246	852	521	223	87	122	321	702	1084	1504		
	CDD50	1378	0	0	1	6	78	257	445	384	183	24	0	0		
	CDD65	136	0	0	0	0	3	16	67	40	10	0	0	0		
	CDH74	1933	0	0	0	2	90	291	888	516	141	5	0	0		
	CDH80	433	0	0	0	0	15	55	233	106	24	0	0	0		
	WSAvg	4.6	4.6	4.8	5.1	6.0	5.7	4.5	3.7	3.5	3.8	4.9	4.7	4.5		
	PrecAvg	31.7	1.3	1.1	1.7	2.7	3.1	3.9	3.5	3.2	3.3	3.6	2.3	1.6		
(15) Precipitation	PrecMax	37.2	3.0	3.2	3.5	9.4	5.6	7.2	7.8	5.8	6.6	8.0	5.0	3.7		
	PrecMin	23.3	0.2	0.2	0.5	0.6	1.8	0.8	1.4	1.5	0.6	0.8	0.5	0.3		
	PrecStd	4.2	0.7	0.7	0.7	2.0	1.0	1.6	1.9	1.3	1.4	1.9	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	45.8	59.3	71.8	82.4	85.3	89.9	87.8	83.5	72.8	58.7	43.3		
		MCWB	35.4	37.8	46.7	53.1	63.2	70.2	70.7	70.7	68.2	59.2	49.8	36.6		
	2%	DB	36.2	38.6	52.0	63.4	76.7	81.1	84.4	82.2	78.8	65.8	52.0	37.3		
		MCWB	33.5	34.1	42.5	47.8	59.2	64.4	68.9	67.3	66.1	55.2	46.1	34.7		
	5%	DB	33.7	35.8	45.7	56.8	70.4	77.3	82.0	79.5	73.2	61.0	46.4	35.9		
		MCWB	31.6	32.6	38.4	44.3	55.7	62.4	67.0	65.3	62.2	52.6	41.9	34.0		
	10%	DB	29.6	31.9	40.9	52.4	64.2	73.1	79.4	76.5	70.1	55.2	43.4	33.8		
		MCWB	28.2	29.3	35.1	42.1	52.8	60.6	65.1	64.1	61.1	48.9	39.3	32.0		
	0.4%	WB	36.7	39.2	50.5	55.4	67.1	72.3	74.9	73.2	71.2	61.8	52.2	38.4		
		MCDB	40.0	43.8	56.1	66.1	76.6	82.9	84.5	82.5	79.3	69.1	56.7	41.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.7	35.2	43.3	49.5	61.3	68.1	71.7	70.4	67.7	57.4	46.3	35.7		
		MCDB	35.7	38.0	49.4	60.5	71.5	76.9	81.5	79.4	74.7	63.1	49.6	37.0		
	5%	WB	30.7	32.1	39.1	46.0	57.7	64.6	69.2	68.0	65.0	53.4	43.1	33.5		
		MCDB	32.6	35.0	45.0	55.2	67.1	72.7	78.8	76.3	71.2	58.6	46.8	35.1		
	10%	WB	27.7	28.9	35.7	42.5	54.3	62.3	67.1	66.0	62.3	50.1	39.3	31.3		
		MCDB	28.8	31.4	40.8	51.2	62.8	71.1	76.1	73.7	67.9	55.3	43.4	33.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	22.6	21.9	21.0	23.0	23.2	23.9	23.5	22.8	18.3	16.2	16.9		
		MCDBR	17.9	23.7	25.9	32.3	33.5	30.4	28.1	28.1	28.2	27.9	22.0	16.5		
		MCWBR	15.1	18.8	17.6	19.3	18.1	16.4	14.2	15.5	17.0	18.3	15.9	13.7		
		MCDBR	17.2	20.2	23.8	29.2	27.3	25.2	25.8	24.5	24.5	23.5	19.7	14.4		
	5% WB	MCWBR	14.8	16.4	16.7	18.5	16.4	15.7	14.9	15.1	16.3	16.7	15.2	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.278	0.310	0.342	0.368	0.378	0.388	0.379	0.358	0.313	0.289	0.266		
	taud		2.440	2.366	2.374	2.385	2.385	2.390	2.395	2.437	2.466	2.560	2.526	2.455		
	Ebn at Noon		272	285	290	289	284	281	276	275	271	270	255	252		
	Edh at Noon		21	28	32	35	36	37	36	33	30	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		402	709	1101	1466	1709	1840	1935	1629	1210	731	407	301		
	RadStd		44	73	101	152	122	128	78	108	73	83	52	30		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page